Task 6: Researching other Logics + LLMs

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Overview

1. I chose to research fuzzy logic. I made slides to offer a more in-depth understanding
2. This deliverable acts as a tutorial and explanation of fuzzy logic and its relevance
3. Fuzzy logic changes traditional 0/1 logic to anything between 0 and 1
4. Fuzzy logic algorithms could be used alongside LLMs to help reduce hallucinations

Slides

The slides cover the concepts in this deliverable in more depth, offer a better visual representation, and include more equations. Please check it out: 🔗

Everyday Use

We use fuzzy logic every day without realizing it. We often measure things on a scale of 0 to 1. For example, I am feeling 0.7 happy today. Fuzzy logic formalizes this by adding rules. Let's define happy as H. Therefore, H = 0.7. This means that not H (or H complement) = 0.3. This means that if I am 0.7 happy, then I am 0.3 not happy. Fuzzy logic models real-world scenarios better than First Order Logic (FOL).

Fuzzy Logic Algorithm

Let's keep going with the happiness example. Many factors go into why I am feeling this way. For example, I got a raise, which makes me 0.8 happy, but I forgot to pack lunch, which makes me 0.6 happy. The average is 0.7 Happy. This is an extremely simplified version of the fuzzy algorithm. In more formal terms, it looks like this:

Factors data → Fuzzified data → Aggregate → Defuzzify

- Factors: raise and lunch: Money from 0 to 1 over the money $ value; hunger from 0 to 1 over the food in my stomach.
- Fuzzified: Graph quantified rules like "a lot of money or not hungry → happy."
- Aggregate: Graph of happiness level as the two factors (money and hunger) change.
- Defuzzify: Locate your level of happiness on the graph

This algorithm is responsible for creating projection models worldwide. For example, weather projection models. You can find more complex examples in the slides.

Relevance in AI and LLMs

The fuzzy logic algorithm is already part of AI, which uses algorithms and data to imitate human experts' instincts. Also known as "expert systems" (Williams, 2020). These types of algorithms can reduce the likelihood of hallucination because they rely on logic and data. We could potentially use such algorithms as a layer on top of LLM neural networks to help reduce hallucinations. Fuzzy logic is well-equipped to handle the complications of natural language.

References

Williams, J. K. (2009). Introduction to fuzzy logic. In S. E. Haupt, A. Pasini, & C. Marzban (Eds.), *Artificial Intelligence Methods in the Environmental Sciences* (pp. 127-149). Springer. https://doi.org/10.1007/978-1-4020-9119-3_6